Open-Source RAG System — Evaluation Report

RBI Document QA · qwen2.5:7b · BAAI/bge-base-en-v1.5 · BAAI/bge-reranker-base · Ollama

1. System Overview

Retrieval-Augmented Generation pipeline over RBI Master Directions corpus. Documents parsed to Markdown, chunked (512 tokens, 64 overlap), indexed with FAISS + BGE embeddings. Optional cross-encoder reranking (bge-reranker-base). Generation and LLM judge use qwen2.5:7b locally via Ollama. All evaluations run on 40-question dataset (26 retrieval-labeled, 8 unanswerable, 6 ambiguous).

2. Retrieval: Baseline vs. Reranking (n=40, same dataset)

Configuration	Recall@5	Recall@10	MRR@10	Retrieval p50	Retrieval p95
Baseline (FAISS only)	0.769	0.885	0.648	—	—
+ bge-reranker-base (×4 candidates)	0.962 (+25pp)	0.962 (+8pp)	0.862 (+33pp)	14.1 s	15.2 s

Source: results/reranking.json (retrieval_only=True, 40 Qs). Reranking retrieves 4× candidates then re-scores with cross-encoder; final top-10 returned.

3. Answer Quality — Full-RAG with LLM Judge (n=40)

Metric	Value	Notes
LLM judge mean correctness	0.765	34 judged (6 skipped: no gold answer)
LLM judge mean faithfulness	0.894	Proportion of claims verifiable in context
Exact-substring correctness	0.115	Naive heuristic; prefer judge score
Judge failures	0	0/34 Ollama calls failed

Source: results/baseline_qwen.json (full_rag, qwen2.5:7b judge).

4. Abstention / Evidence-Grounded Refusal

Run	Abstention Recall	Abstention Precision	Hallucination Rate	n
Targeted (unanswerable-only)	1.000 (8/8)	1.000 (0 FP)	0.000	8
Full 40-q run (corrected detector)	1.000 (8/8)	0.889 (1 FP)	0.000	40
Full 40-q run (original detector)	0.000	—	1.000	40

Original run: model produced semantically-correct but lexically-divergent refusals that escaped 9-pattern detector. Fix: 5 additional patterns added to is_refusal_response() in generation.py; all 8 now detected. 1 false positive in full run (answerable Q answered cautiously). Source: results/abstention_targeted.json, results/abstention_qwen.json.

5. LLM Judge Validation — Cohen's Kappa

Dataset	n pairs	TP	TN	FP	FN	κ
Full 40-q (corrected detector)	34	8	26	0	0	1.000
Full 40-q (original detector)	34	0	26	0	8	0.000
Targeted 8-q (unanswerable only)	8	8	0	0	0	1.000

Human labels: 'answerable' field from eval/qa_pairs.jsonl (genuine, not fabricated). Model decision: is_refusal flag. Kappa=1.0 with corrected detector; kappa=0.0 with original detector because the model never refused (all FN).

6. Latency (Measured, n=40 per row)

Component	mean	p50	p95	min	max	Source
Retrieval + reranking (old run)	14.0 s	14.1 s	15.2 s	12.2 s	15.6 s	reranking.json
Retrieval + reranking (qwen run)	18.6 s	16.7 s	30.4 s	14.5 s	37.0 s	reranking_qwen.json
Full RAG baseline (no judge)	86.8 s	83.4 s	121.7 s	12.1 s	136.6 s	baseline.json
Full RAG + LLM judge	102.5 s	91.3 s	174.7 s	79.3 s	208.5 s	baseline_qwen.json

Generation latency ~84 s mean (full-RAG minus retrieval-only). Dominated by qwen2.5:7b inference on local CPU/GPU. No isolated generation-only run available; estimate is derived. Retrieval latency is the measured retrieval+reranking step.

7. Best Validated Configuration

Dimension	Best Result	Configuration
Retrieval	Recall@5=0.962, MRR@10=0.862	FAISS + BGE + bge-reranker-base
Answer correctness	0.765 (LLM judge)	qwen2.5:7b, temp=0, grounded prompt
Answer faithfulness	0.894 (LLM judge)	qwen2.5:7b, temp=0
Abstention (targeted)	Recall=1.0, Prec=1.0	Strict refusal rule + expanded detector
Abstention (full run)	Recall=1.0, Prec=0.889	After pattern fix

8. Limitations & Honest Caveats

• Abstention recall=1.0 on the targeted 8-question run; full-run precision=0.889 (1 false refusal).
• Exact-substring correctness (0.115) is a conservative floor; LLM judge score (0.765) is the primary quality signal.
• Latency 83–102 s per question on local hardware; unsuitable for interactive use without GPU acceleration.
• LLM judge uses same model (qwen2.5:7b) as generator — potential self-consistency bias; no external oracle.
• Cohen's kappa=1.0 after detector fix; before fix kappa=0.0 due to detection failure, not generation failure.
• No git remote configured; tag v1.0 is local only. Push not verified.
• 40-question dataset is manually curated; test set size limits statistical power.

Tests: 63/63 pass (pytest -q) · Lint: ruff check (BLE001 only in pre-existing app.py / test_smoke.py) · Demo: streamlit run app.py · Corpus: RBI Master Directions (parsed Markdown, FAISS index)